Chapter 2: Radical Functions

MULTIPLE CHOICE

1. Compared to the graph of the base function $f(x) = \sqrt{x}$, the graph of the function $g(x) = \sqrt{x} - 2$ is translated

A 2 units up
B 2 units to the right
C 2 units to the left
D 2 units down

ANS: D DIF: Easy OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: vertical translation

2. Compared to the graph of the base function $f(x) = \sqrt{x}$, the graph of the function $g(x) + 8 = \sqrt{x}$ is translated

A 8 units to the left
B 8 units up
C 8 units to the right
D 8 units down

ANS: D DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: vertical translation

3. Compared to the graph of the base function $f(x) = \sqrt{x}$, the graph of the function $g(x) = \sqrt{x-5}$ is translated

A 5 units down
B 5 units left
C 5 units right
D 5 units up

ANS: C DIF: Easy OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: horizontal translation

4. Compared to the graph of the base function $f(x) = \sqrt{x}$, the graph of the function $g(x) = \sqrt{5x}$ is

A compressed by a factor of $\frac{1}{5}$ and not reflected
B stretched by a factor of 5 and reflected in the y-axis
C compressed by a factor of $\frac{1}{5}$ and reflected in the y-axis
D stretched by a factor of 5 and not reflected

ANS: A DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: horizontal stretch

5. When $b < 0$, the function $g(x) = \sqrt{bx}$ has what relationship to the base function $f(x) = \sqrt{x}$?

A $f(x)$ is stretched horizontally by a factor of 1/|b|
B $f(x)$ is stretched horizontally by a factor of 1/|b| and reflected in the y-axis
C $f(x)$ is stretched vertically by a factor of |b|
D $f(x)$ is stretched vertically by a factor of |b| and reflected in the x-axis

ANS: B DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: horizontal stretch | reflection

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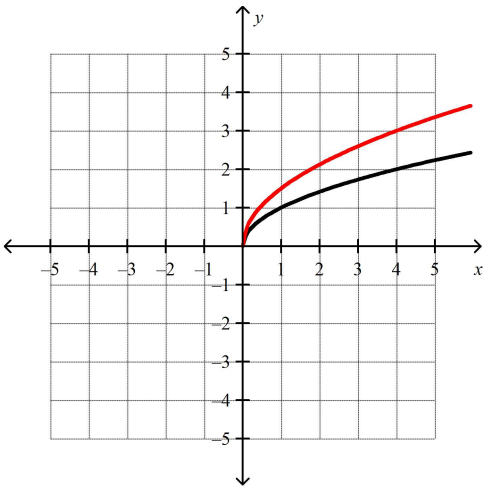
6. When the graph of $y = \sqrt{x}$ is vertically stretched by a factor of 3, the point (9, –10) will be translated to the point

A (–10, 9)
B (3, –30)
C (–30, 9)
D (9, –30)

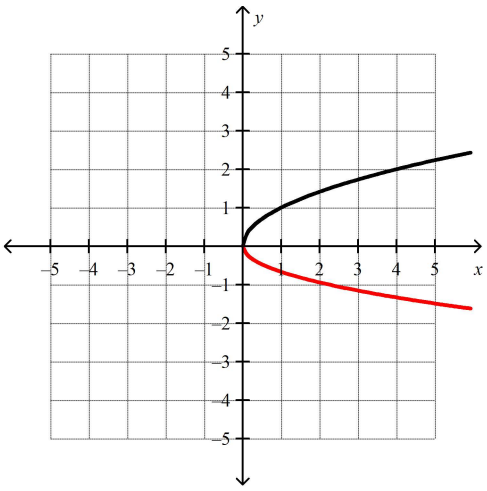
ANS: D DIF: Easy OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: vertical stretch

7. Which of the graphs shown below represents the base function $f(x) = \sqrt{x}$ (in black) and the stretched function $g(x) = -\frac{3}{2}\sqrt{x}$ (in red)?

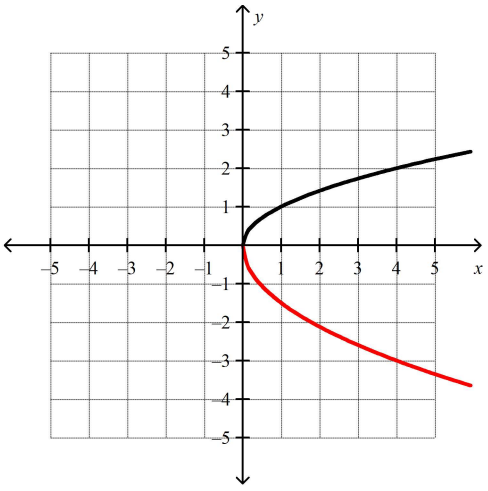
A



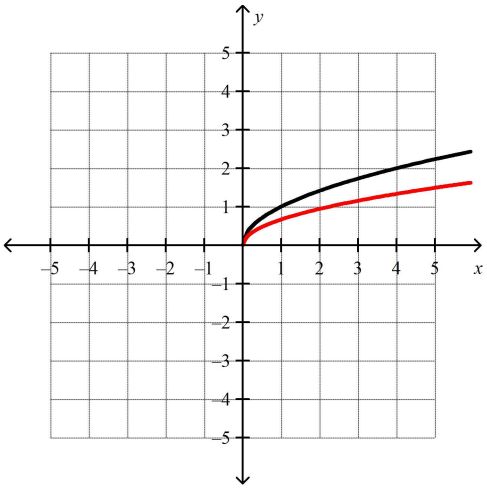
C



B



D



ANS: B DIF: Easy OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | vertical stretch | reflection

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8. Given the function $f(x) = \sqrt{x-h} + k$ with a domain of $\{x | x \geq -5, x \in R\}$ and a range of $\{y | y \geq 8, y \in R\}$, which of the following best describes the vertical and horizontal translations with respect to the graph of $f(x) = \sqrt{x}$?

A 5 units to the left and 8 units up
B 8 units to the left and 5 units down
C 8 units to the left and 5 units up
D 5 units to the left and 8 units down

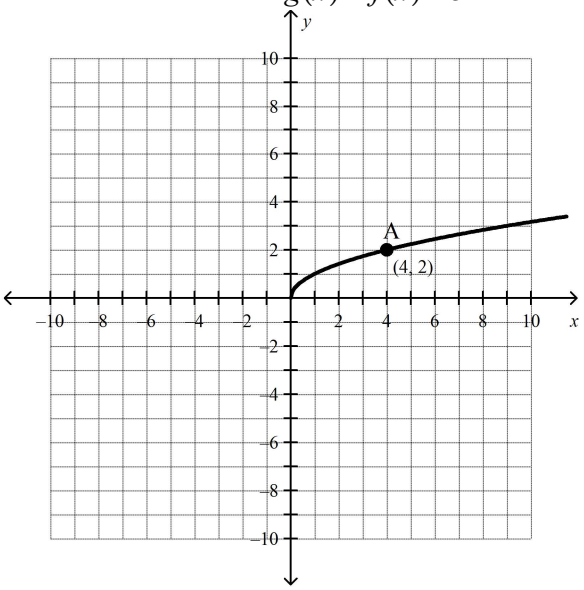
ANS: A DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: horizontal translation | vertical translation

9. Compared to the graph of the base function $f(x) = \sqrt{x}$, the graph of the function $g(x) = \sqrt{x+8} - 4$ is translated

A 4 units to the right and 8 units up
B 4 units to the left and 8 units down
C 8 units to the left and 4 units down
D 8 units to the right and 4 units up

ANS: C DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: horizontal translation | vertical translation

10. Given the graph of $f(x)$ shown below, what are the coordinates of point A′ if the equation of the transformed function is $g(x) = f(x) + 5$?

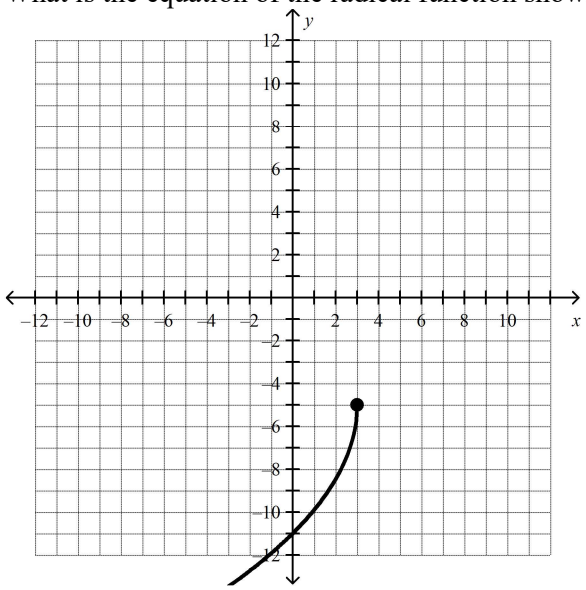


A (–1, 7)
B (4, 7)
C (4, – 3)
D (9, 7)

ANS: B DIF: Easy OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: vertical translation | graph

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11. What is the equation of the radical function shown in the graph below?



A $f(x) = -2\sqrt{\frac{1}{-3}(x-5)} - 3$

B $f(x) = -2\sqrt{-3(x+3)} - 5$

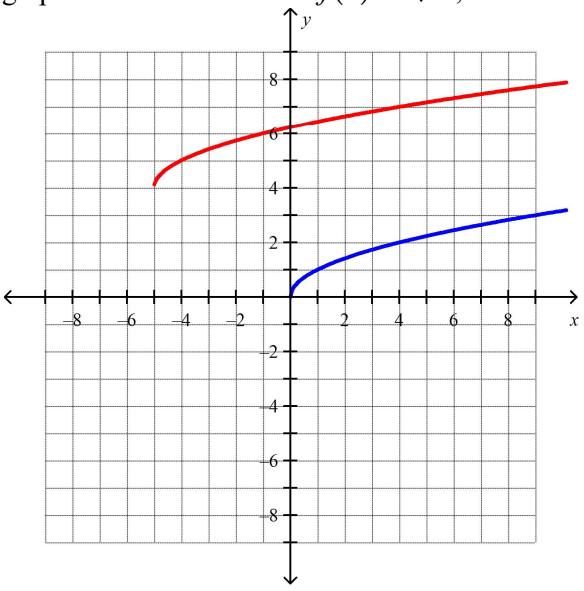
C $f(x) = -2\sqrt{-3(x-3)} - 5$

D $f(x) = -2\sqrt{\frac{1}{-3}(x-3)} - 5$

ANS: C DIF: Difficult OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: horizontal translation | vertical translation | vertical stretch | horizontal stretch | graph | reflection

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12. What is the equation of the transformed function, $g(x)$, after the transformations are applied to the graph of the base function $f(x) = \sqrt{x}$, shown in blue, to obtain the graph of $g(x)$, shown in red?



A $g(x) + 4 = \sqrt{x+4}$

B $g(x) = \sqrt{x+4} + 5$

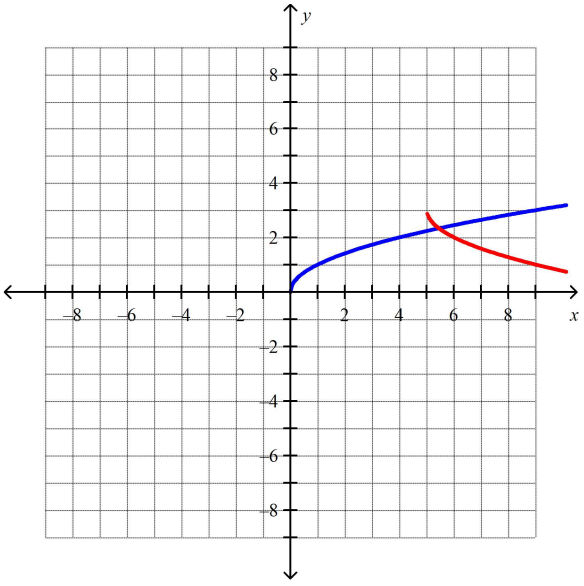
C $g(x) + 5 = \sqrt{x+4}$

D $g(x) = \sqrt{x+5} + 4$

ANS: D DIF: Difficult OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: horizontal translation | vertical translation

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13. In the following graph, what transformations must be applied to the blue curve to obtain the red curve?



A a reflection in the *x*-axis, a vertical translation 5 units up, and a horizontal translation 3 units to the right

B a reflection in the *x*-axis, a vertical translation 5 units down, and a horizontal translation 3 units to the right

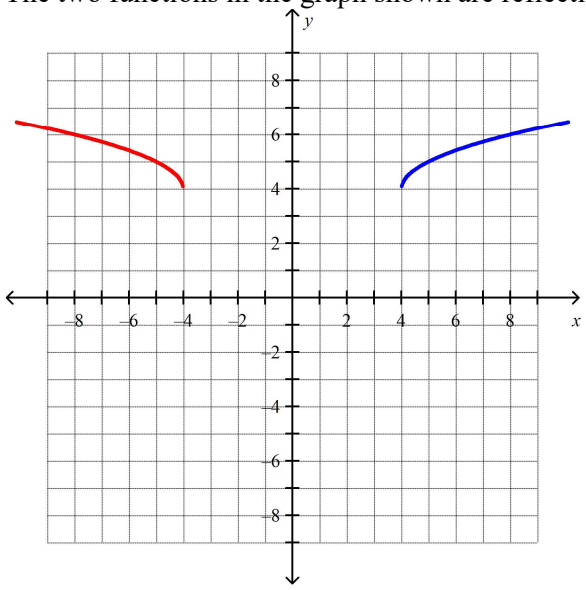
C a reflection in the *x*-axis, a vertical translation 3 units up, and a horizontal translation 5 units to the left

D a reflection in the *x*-axis, a vertical translation 3 units up, and a horizontal translation 5 units to the right

ANS: D DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | horizontal translation | vertical translation | reflection

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14. The two functions in the graph shown are reflections of each other. Select the type of reflection(s).



A a reflection in the y-axis
B a reflection in the x-axis and the y-axis
C a reflection in the line $y = x$
D a reflection in the x-axis

ANS: A DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: reflection

15. Given the function $f(x) = -\sqrt{x} - 4$, which of the following is a restriction on the inverse function, $f^{-1}(x)$?

A $x \geq 8$
B $x \geq 4$
C $x \leq -4$
D $x \geq -4$

ANS: C DIF: Difficult OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: inverse of a radical function | range

16. Which point on the graph $y = f(x)$ does not exist on the graph of $y = \sqrt{f(x)}$?

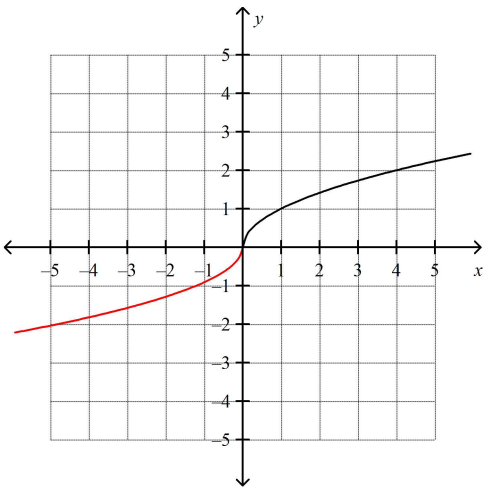
A $(3,4)$
B $(-4,4)$
C $(-8,-3)$
D $(0.6,2.7)$

ANS: C DIF: Easy OBJ: Section 2.2 NAT: RF13
TOP: Square Root of a Function KEY: domain | range

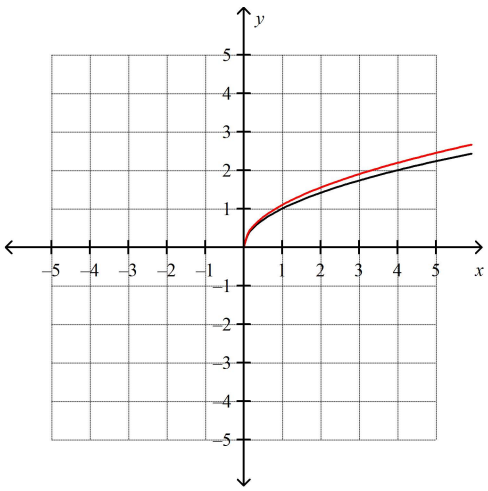
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17. Which of the graphs shown below represents the base function $f(x) = \sqrt{x}$ (in black) and the stretched function $g(x) = \sqrt{5/6x}$ (in red)?

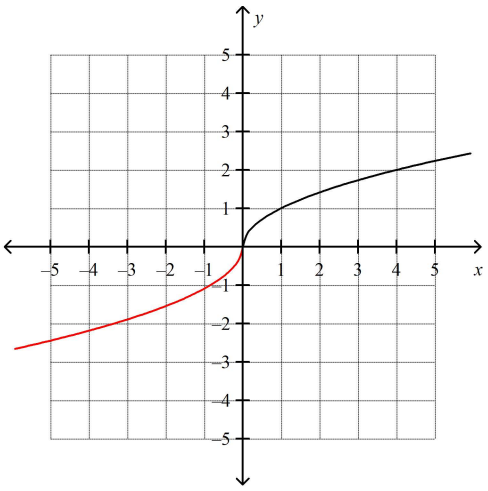
A



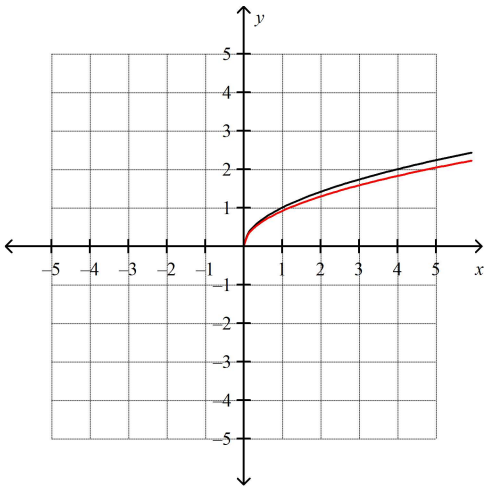
C



B



D



ANS: D DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | horizontal stretch | reflection

18. What are the coordinates of the invariant point(s) when the function $y = \sqrt{x} - 3$ is reflected in the y-axis?

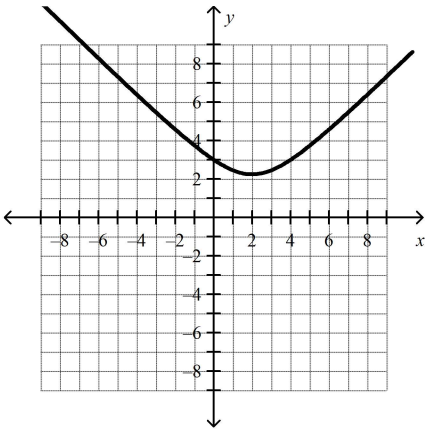
A (9, –3)
B (–3, 0) and (9, 0)
C (0, –3)
D (0, 9)

ANS: C DIF: Difficult OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: invariant points

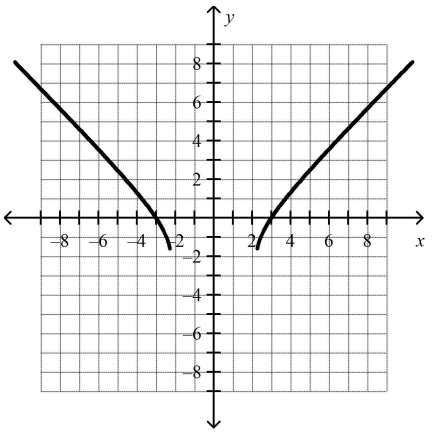
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19. Which is the graph of the square root of the function $f(x) = (x-5)^2 - 2$?

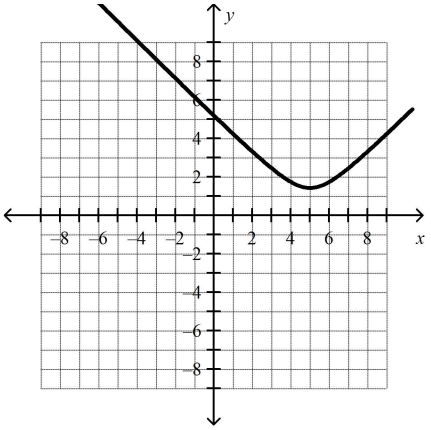
A



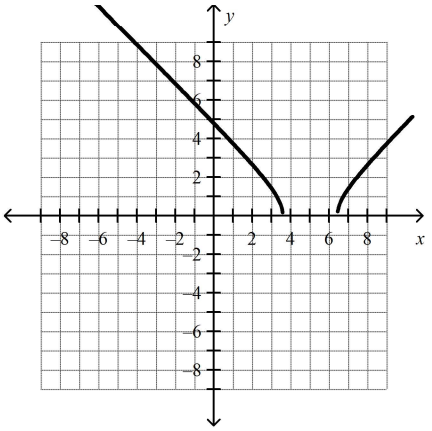
C



B



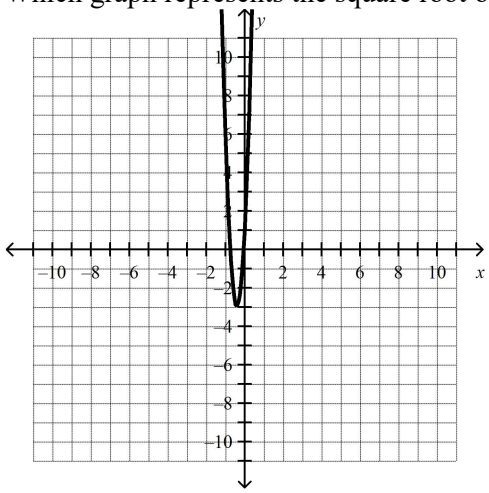
D



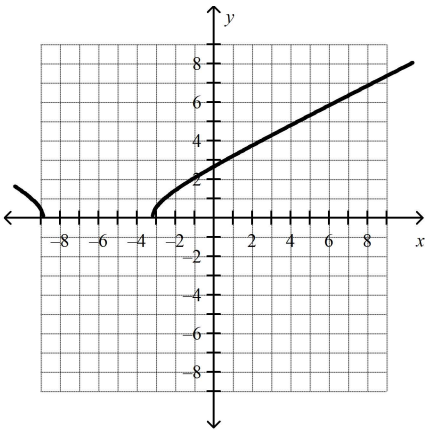
ANS: D DIF: Average OBJ: Section 2.2 NAT: RF13
TOP: Square Root of a Function KEY: graph

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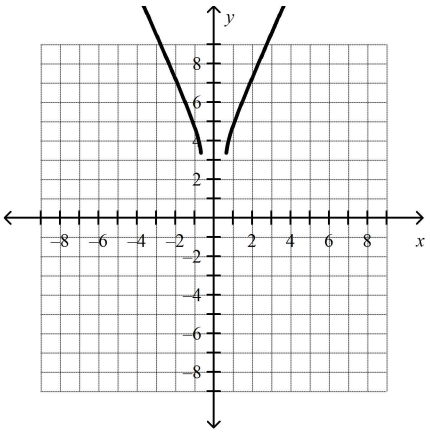
20. Which graph represents the square root of the function shown in the graph?



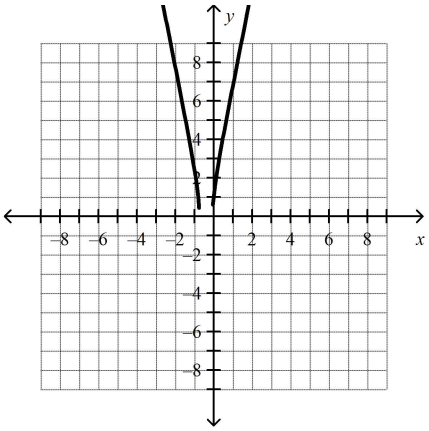
A



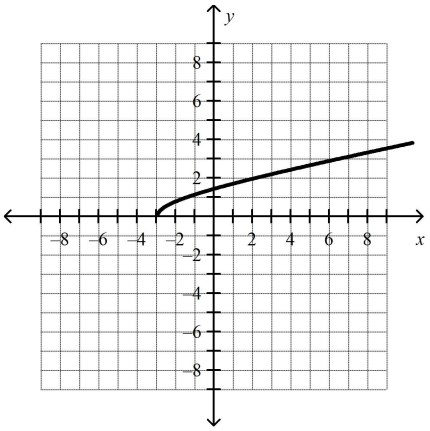
C



B



D



ANS: B DIF: Average OBJ: Section 2.2 NAT: RF13
TOP: Square Root of a Function KEY: graph

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21. Which of the following functions is the correct inverse for the function $f(x) = \sqrt{x-2}$, $\{x \mid x \geq 0, x \in R\}$?

A $f^{-1}(x) = \sqrt{x+2}$

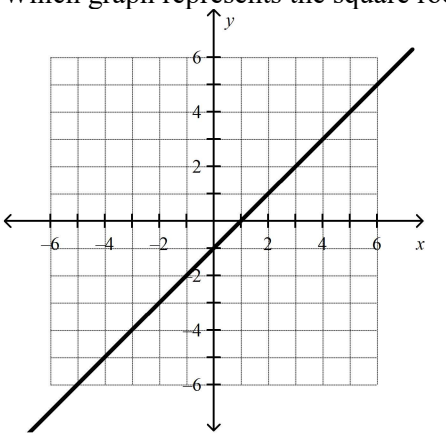
B $f^{-1}(x) = \sqrt{x} + 2$

C $f^{-1}(x) = x^2 + 2$

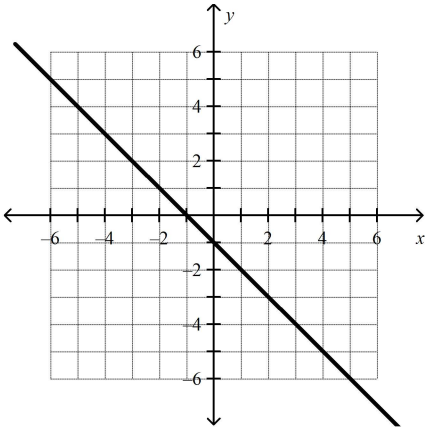
D $f^{-1}(x) = (x-2)^2$

ANS: C DIF: Easy OBJ: Section 2.1 | Section 2.2
NAT: RF13 TOP: Radical Functions and Transformations | Square Root of a Function
KEY: inverse of a radical function

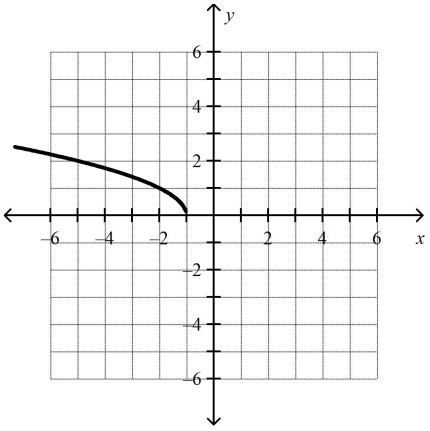
22. Which graph represents the square root of the graph shown?



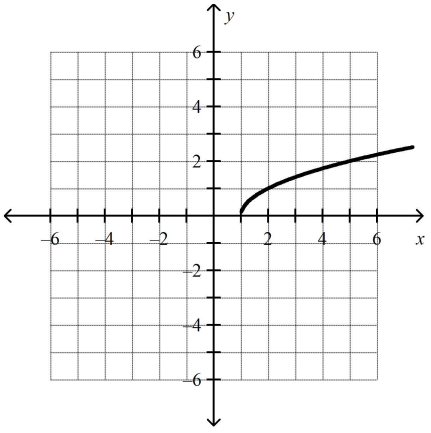
A



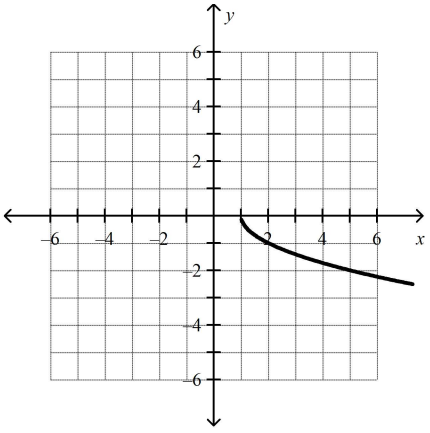
C



B



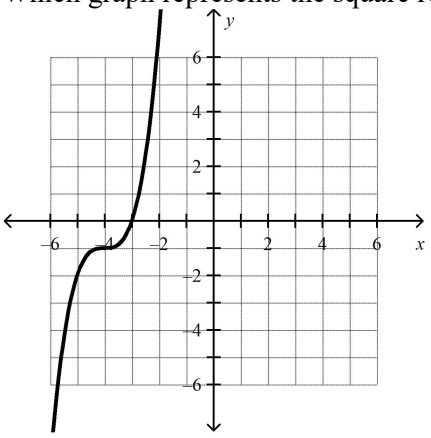
D



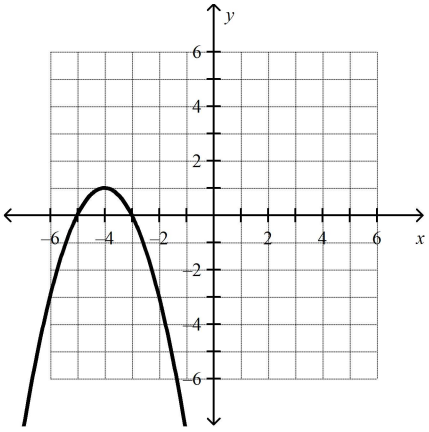
ANS: B DIF: Easy OBJ: Section 2.2 NAT: RF13
TOP: Square Root of a Function KEY: graph | square root of a function

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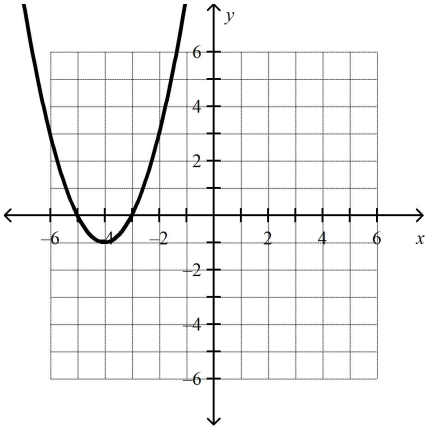
23. Which graph represents the square root of the graph shown?



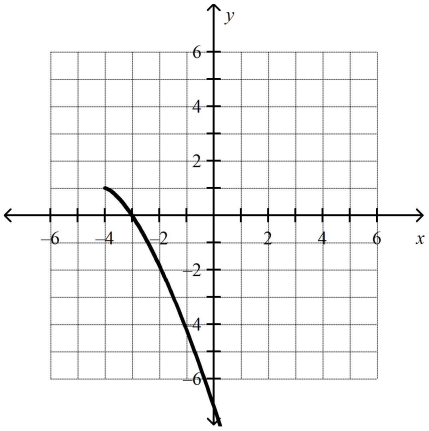
A



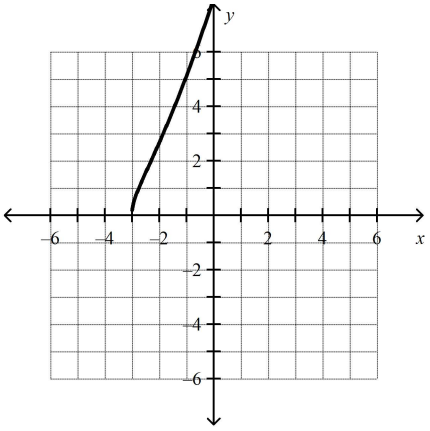
C



B



D

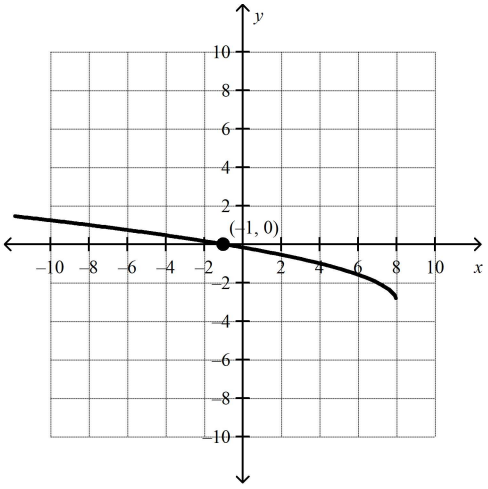


ANS: D DIF: Average OBJ: Section 2.2 NAT: RF13
TOP: Square Root of a Function KEY: graph | square root | of a function

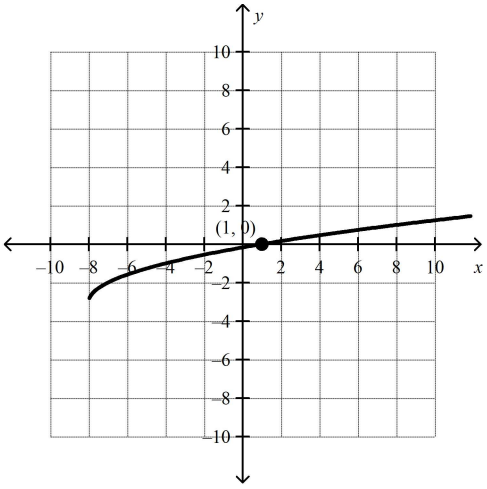
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24. Which graph shows the graphical solution to the radical equation $0 = \sqrt{x+8} - 3$?

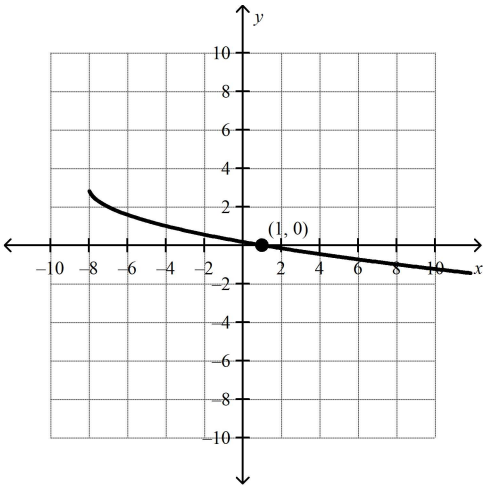
A



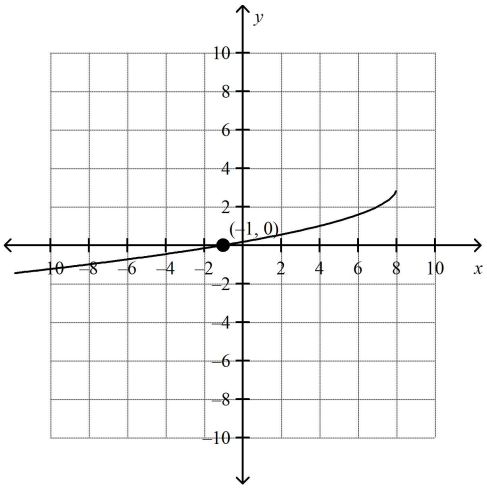
C



B



D

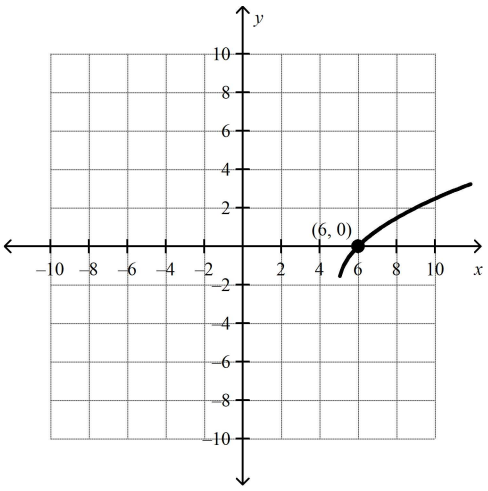


ANS: C DIF: Average OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: graphical solution

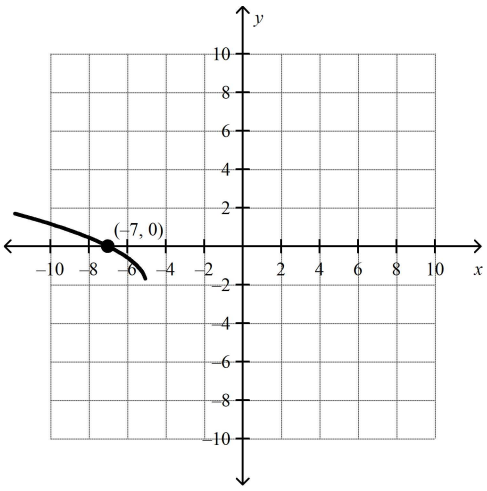
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25. Which graph shows the graphical solution to the radical equation $0 = 2\sqrt{(x-5)} - 2$?

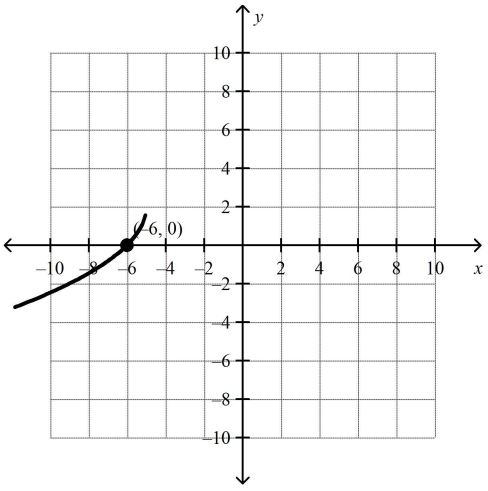
A



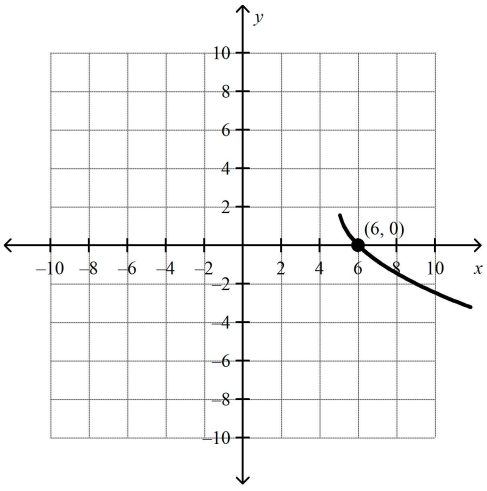
C



B



D

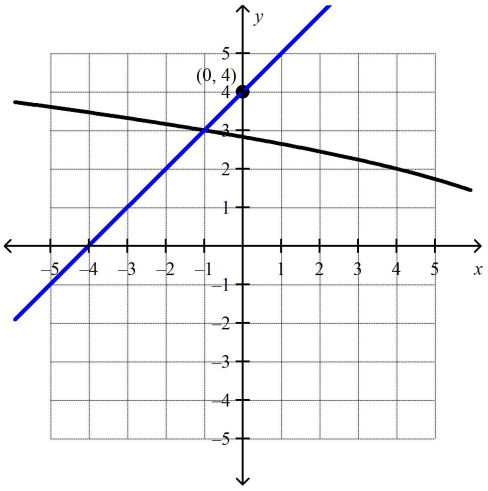


ANS: A DIF: Average OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: graphical solution

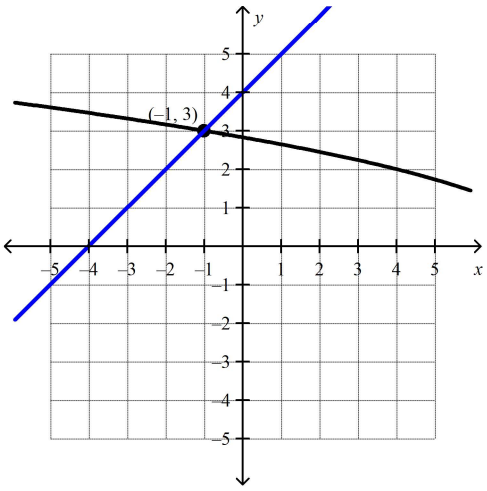
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26. Which graph shows the solution to the radical equation $\sqrt{8-x} = x+4$?

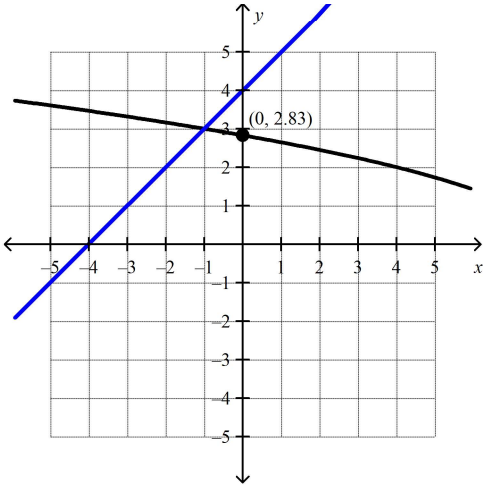
A



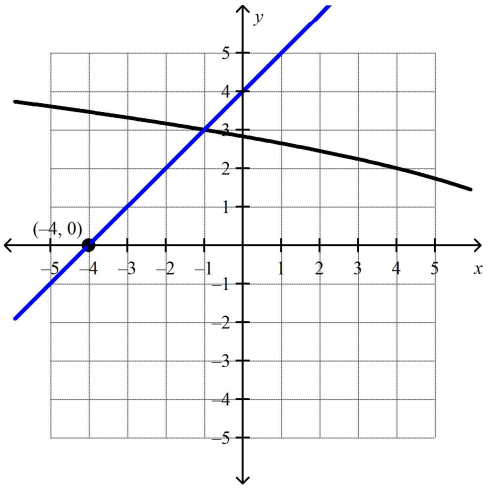
C



B



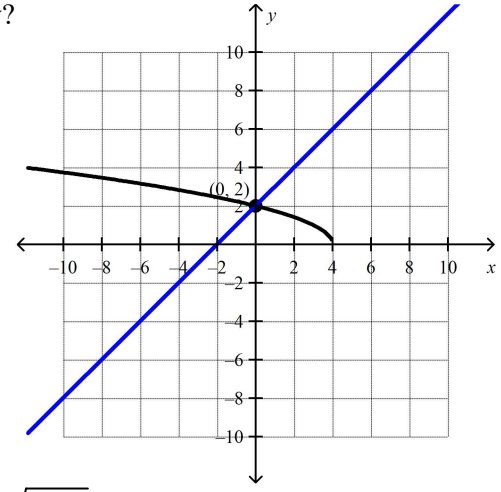
D



ANS: C DIF: Easy OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: graphical solution

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27. Which radical equation can be solved using the graph shown below?



A $-\sqrt{4-x}=x+2$

B $\sqrt{4-x}=x+2$

C $x+2=-\sqrt{4+x}$

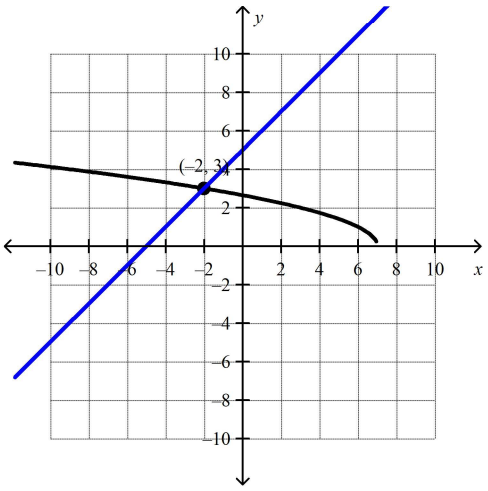
D $\sqrt{4+x}=x+2$

ANS: B DIF: Easy OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: graphical solution

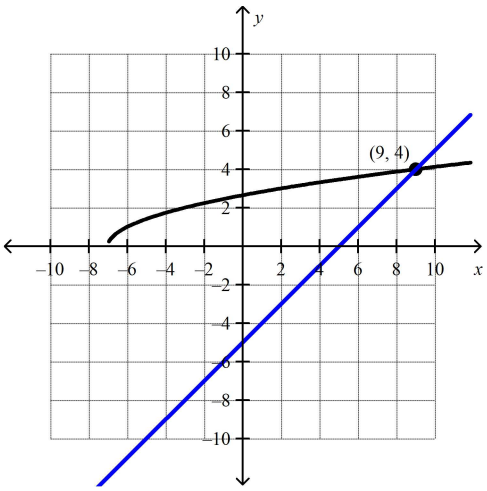
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28. Which graph shows the solution to the radical equation $-\sqrt{7-x} = -x-5$?

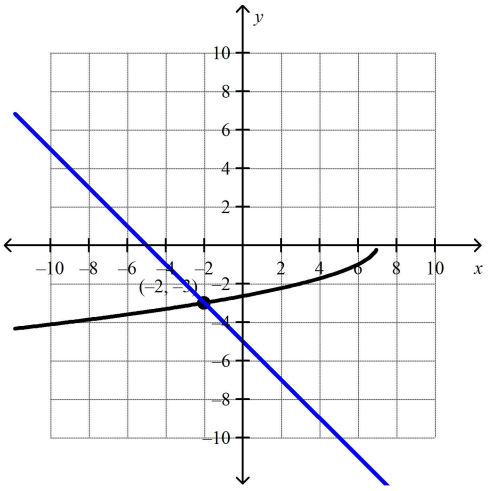
A



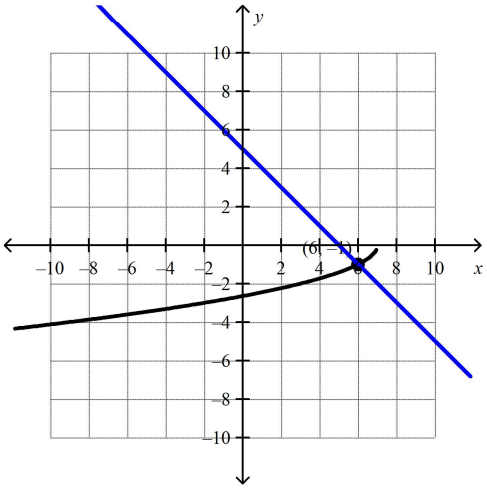
C



B



D



ANS: B DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: graphical solution

29. What is the solution to the radical equation $0 = \sqrt{x+9} - 3$?

A 18
B 36
C –18
D 0

ANS: D DIF: Average OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

30. What is the solution to the radical equation $0 = 2\sqrt{2(x+4)} - 8$?

A –4
B 12
C 4
D 128

ANS: C DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

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31. What is the solution to the radical equation $-10 = -10\sqrt{(x+7)}$?

A 6 **C** 7
B 8 **D** –6

ANS: D DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

32. When solving the equation $\sqrt{x+8} = x+6$, which values must be checked for extraneous roots?

A –4 and 7 **C** 8 and 6
B 7 and –7 **D** –4 and –7

ANS: D DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically
KEY: algebraic solution | extraneous roots

33. Which equation of a radical function would have the following domain and range?
$\{x|x \geq -6, x \in R\}$; $\{y|y \geq 10, x \in R\}$

A $y = \sqrt{x-10} + 6$ **C** $y = \sqrt{x+6} - 10$
B $y = \sqrt{x+6} + 10$ **D** $y = \sqrt{x+10} + 6$

ANS: B DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: domain | range

COMPLETION

1. The variable k in the function $f(x) = a\sqrt{b(x-h)} + k$ represents a _____.

ANS: vertical translation

DIF: Easy OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: translations | stretches

2. The graph of $g(x) = \sqrt{f(x+8)}$ is the graph of $f(x)$ moved 8 units to the _____.

ANS: left

DIF: Easy OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: horizontal translation | vertical translation

3. If the value of $f(x)$ satisfies f(x) > 1, then the graph of $y = \sqrt{f(x)}$ _____.

ANS: is below the graph of y = f(x)

DIF: Difficult OBJ: Section 2.2 NAT: RF13
TOP: Radical Functions and Transformations KEY: relative location

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4. If $f(x) = -9 + 6x^2$, the range of $y = \sqrt{f(x)}$ is _____.

ANS: $\{y | y \geq \sqrt{3/2}, y \in R\}$

DIF: Average OBJ: Section 2.2 NAT: RF13 TOP: Square Root of a Function
KEY: range | square root

5. When solving the equation $0 = -10 + \sqrt{-3x + 6} - 6$ algebraically, the restriction on the variable is _____.

ANS: $x \leq 2$

DIF: Average OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

6. The solution to the equation $0 = -4 + 2\sqrt{x + 19}$ is _____.

ANS: (–15, 0)

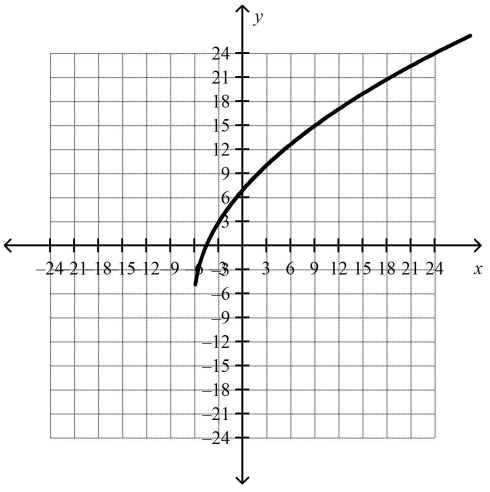
DIF: Average OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

MATCHING

Match the functions to their corresponding graphs.

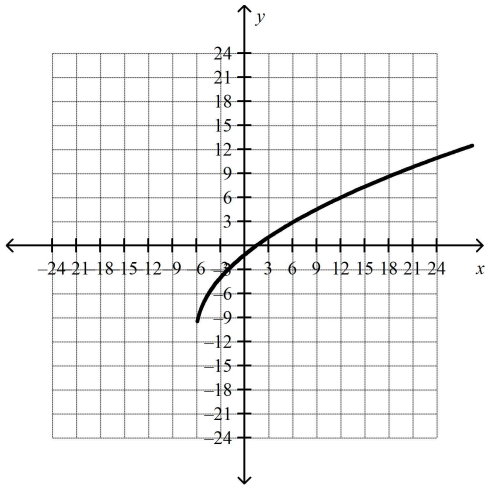
A $f(x) = 4\sqrt{2(x-7)} + 6$
B $f(x) = 2\sqrt{4(x-7)} + 6$
C $f(x) = 2\sqrt{4(x+6)} - 7$
D $f(x) = 4\sqrt{2(x+6)} - 7$
E $f(x) = 4\sqrt{2(x-7)} - 11$
F $f(x) = 2\sqrt{4(x+6)} - 11$

1.

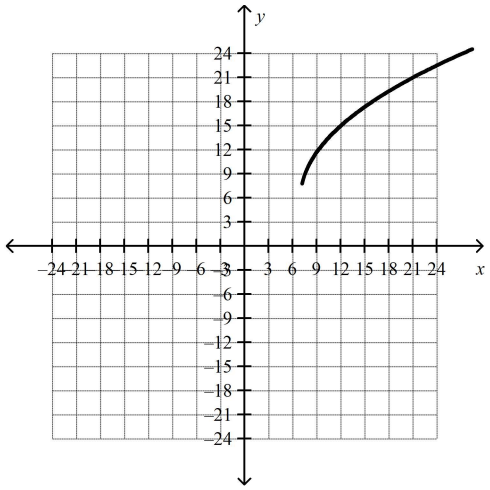


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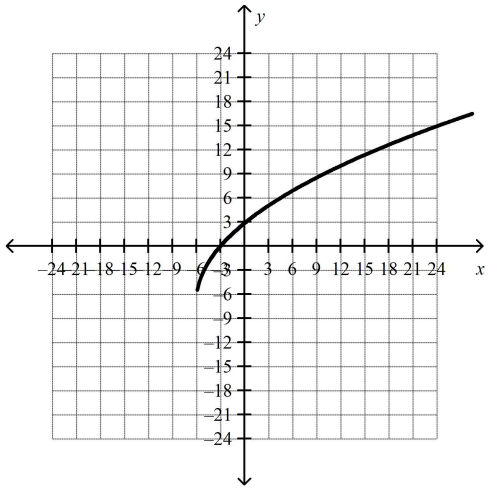
2.



3.

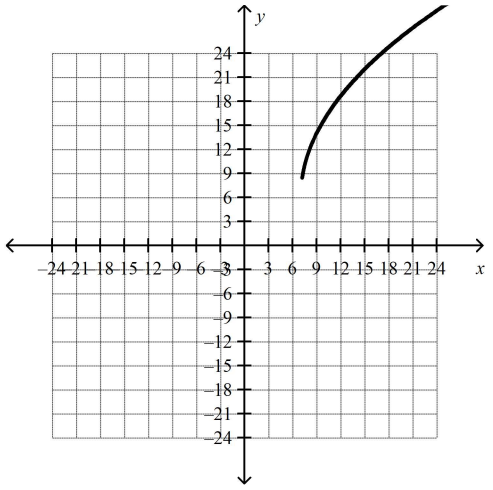


4.



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5.



1. ANS: D DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | transformations | translations | vertical translation | horizontal translation

2. ANS: F DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | transformations | translations | vertical translation | horizontal translation

3. ANS: B DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | transformations | translations | vertical translation | horizontal translation

4. ANS: C DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | transformations | translations | vertical translation | horizontal translation

5. ANS: A DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | transformations | translations | vertical translation | horizontal translation

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SHORT ANSWER

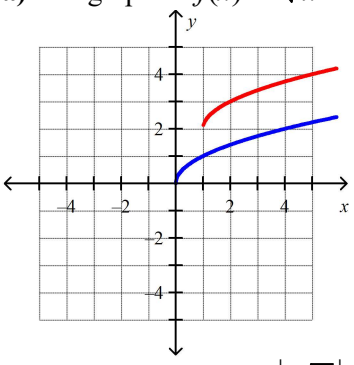
1. Create a graph of $g(x) = f(x-1)+2$ for each base function given, using transformations.

a) $f(x) = \sqrt{x}$

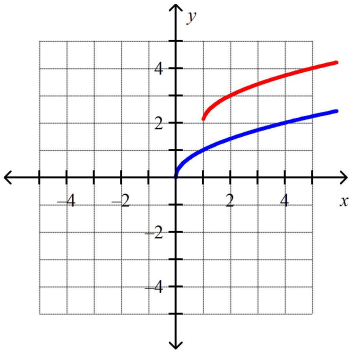
b) $f(x) = \left|\sqrt{x}\right|$

ANS:

a) The graph of $f(x) = \sqrt{x}$ is shown in blue and the graph of $g(x) = \sqrt{x-1}+2$ is shown in red.



b) The graph of $f(x) = \left|\sqrt{x}\right|$ is shown in blue and the graph of $g(x) = \left|\sqrt{x-1}\right|+2$ is shown in red.



DIF: Average OBJ: Section 2.1 NAT: RF13 | RF13
TOP: Radical Functions and Transformations
KEY: translations | vertical translation | horizontal translation

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2. Determine the equation of each radical function, which has been transformed from $f(x) = \sqrt{x}$ by the given translations.

a) vertical stretch by a factor of 5, then a horizontal translation of 6 units right

b) horizontal stretch by a factor of $\frac{1}{6}$, then a vertical translation of 4 units down

c) horizontal reflection in the *y*-axis, then a vertical translation of 9 units up and horizontal translation of 2 units right

d) horizontal stretch by a factor of $\frac{1}{3}$, vertical reflection in the *x*-axis, and vertical stretch by a factor of $\frac{2}{3}$

ANS:

Substitute values into the general equation $g(x) = a\sqrt{b(x-h)} + k$.

a) $g(x) = 5\sqrt{x-6}$

b) $g(x) = \sqrt{6x} - 4$

c) $g(x) = \sqrt{-(x-2)} + 9$

or

$g(x) = \sqrt{-x+2} + 9$

d) $g(x) = -\frac{2}{3}\sqrt{\frac{1}{\frac{1}{3}}x}$

or

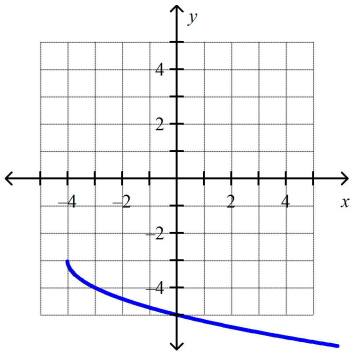
$g(x) = -\frac{2}{3}\sqrt{3x}$

DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: transformations

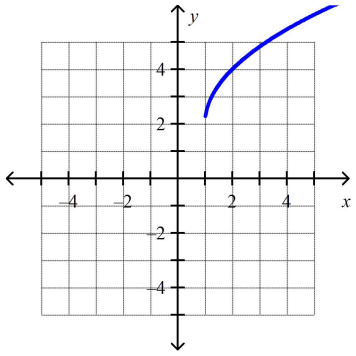
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3. For each graph, write the equation of a radical function of the form $y = a\sqrt{b(x-h)} + k$.

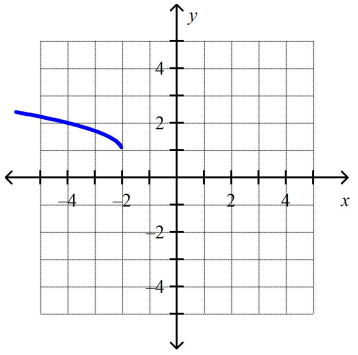
a)



b)



c)



ANS:

a) $y = -\sqrt{x+4} - 3$

b) $y = 2\sqrt{x-1} + 2$

c) $y = \frac{1}{2}\sqrt{-2(x+2)} + 1$

DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: graph | transformations

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4. Determine the domain and range of each function.

a) $f(x) = \sqrt{-x} + \frac{1}{2}$

b) $f(x) = 2\sqrt{\frac{2}{3}(x-2)} + 1$

c) $y + 2 = 2\sqrt{2(x-1)}$

ANS:

a) $\{x | x \leq 0, x \in R\}, \{y | y \geq \frac{1}{2}, y \in R\}$

b) $\{x | x \geq 2, x \in R\}, \{y | y \geq 1, y \in R\}$

c) $\{x | x \geq 1, x \in R\}, \{y | y \geq -2, y \in R\}$

DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: domain | range

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5. **a)** Sketch the graph of each function $f(x)$, and then sketch the graph of $g(x) = \sqrt{f(x)}$.

i) $f(x) = x$

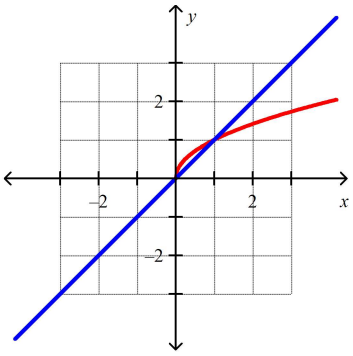
ii) $f(x) = x^2 - 4$

iii) $f(x) = -0.5x + 2$

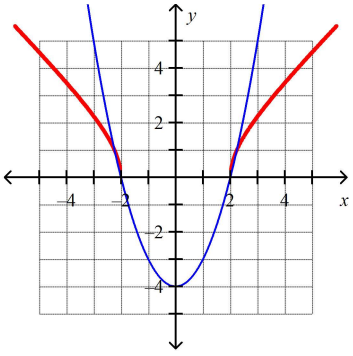
b) State the domain and range of each $f(x)$ and $g(x)$.

ANS:

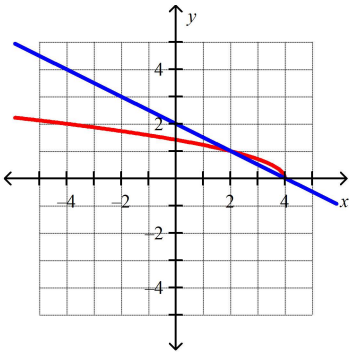
a) i) The graph of $f(x) = x$ is shown in blue and the graph of $g(x) = \sqrt{f(x)}$ is shown in red.



ii) The graph of $f(x) = x^2 - 4$ is shown in blue and the graph of $g(x) = \sqrt{f(x)}$ is shown in red.



iii) The graph of $f(x) = -0.5x + 2$ is shown in blue and the graph of $g(x) = \sqrt{f(x)}$ is shown in red.



b) i) $f(x)$: $\{x | x \in R\}, \{y | y \in R\}$

$g(x)$: $\{x | x \geq 0, x \in R\}, \{y | y \geq 0, y \in R\}$

ii) $f(x)$: $\{x | x \in R\}, \{y | y \geq -4, y \in R\}$

$g(x)$: $\{x | x \geq 2, x \leq -2, x \in R\}, \{y | y \geq 0, y \in R\}$

iii) $f(x)$: $\{x | x \in R\}, \{y | y \in R\}$

$g(x)$: $\{x | x \leq 4, x \in R\}, \{y | y \geq 0, y \in R\}$

DIF: Average OBJ: Section 2.2 NAT: RF13 TOP: Square Root of a Function

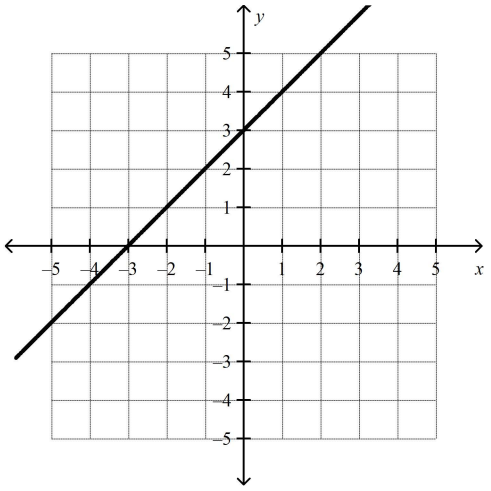
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KEY: graph | domain | range | square root of a function

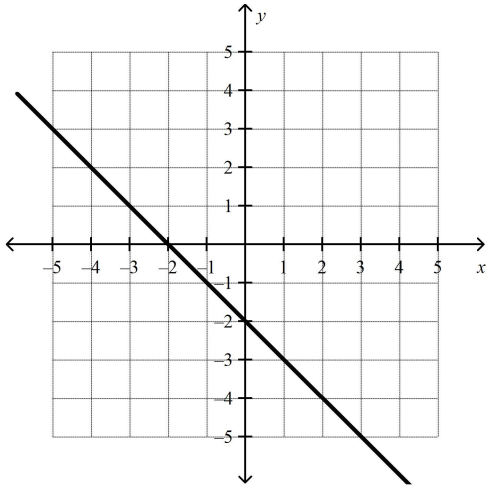
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6. Using each graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.

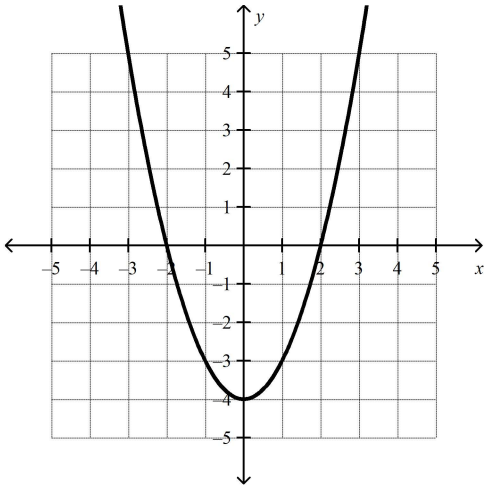
a)



b)



c)

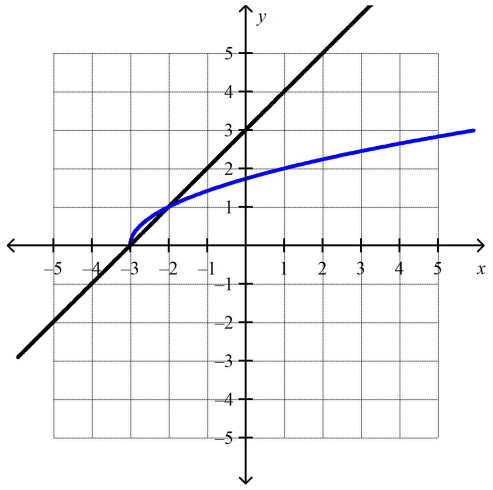


ANS:

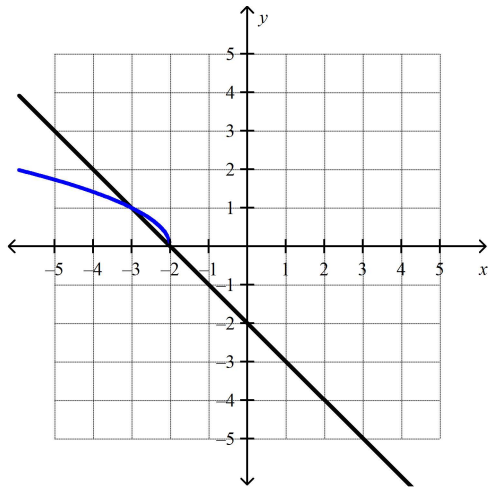
The graph of $y = f(x)$ is shown in black, and the graph of $y = \sqrt{f(x)}$ is shown in blue.

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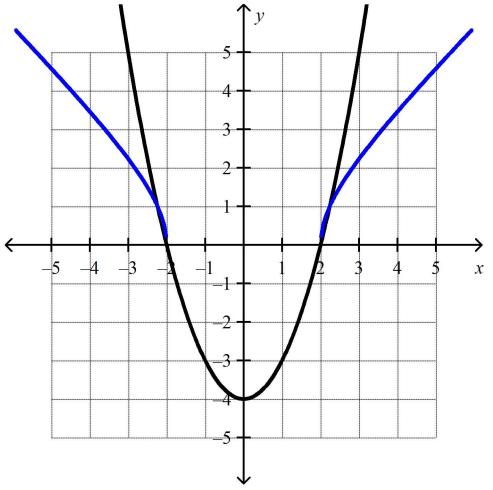
a)



b)



c)



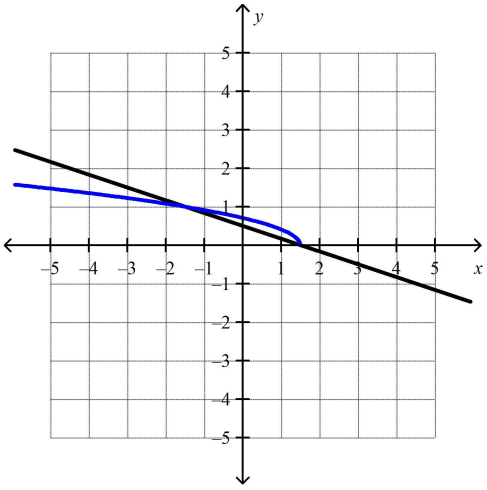
DIF: Average OBJ: Section 2.2 NAT: RF13 TOP: Square Root of a Function
KEY: graph | square root of a function

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7. Sketch the graph of $f(x) = \dfrac{-2x+3}{6}$ and use it to sketch the graph of $y = \sqrt{f(x)}$.

ANS:

The graph of $y = f(x)$ is shown in black, and the graph of $y = \sqrt{f(x)}$ is shown in blue.



DIF: Difficult OBJ: Section 2.2 NAT: RF13 TOP: Square Root of a Function
KEY: graph | square root of a function

8. Solve the equation $3\sqrt{2x+4}+9=12$, $x \geq -2$, algebraically.

ANS:

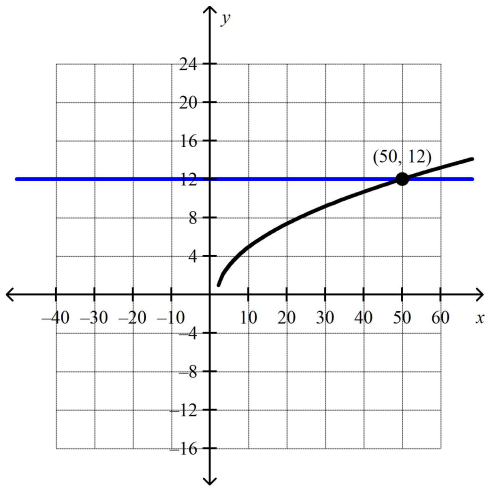
$$3\sqrt{2x+4}+9=12$$
$$3\sqrt{2x+4}=3$$
$$\sqrt{2x+4}=1$$
$$2x+4=1$$
$$2x=-3$$
$$x=-\frac{3}{2}$$

DIF: Average OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

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9. Solve the equation $\sqrt{3x-6} = 12$ graphically.

ANS:



DIF: Average OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: graphical solution

10. Jim states that the equations $\sqrt{x^2} = 25$ and $\left(\sqrt{x}\right)^2 = 25$ have the same solution. Is he correct? Justify your reasoning.

ANS:

No, Jim is not correct. $\sqrt{x^2} = 25$ has two possible solutions of ±25, and $\left(\sqrt{x}\right)^2 = 25$ has only one solution, +25.

DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

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11. A student designs a special container as part of an egg drop experiment. She believes that the container can withstand a fall as long as the speed of the container does not exceed 80 ft/s. She uses the equation $v = \sqrt{(v_0)^2 + 2ad}$ to model the velocity, v, in feet per second, as a function of constant acceleration, a, in feet per second squared and the drop distance, d, in feet. Assuming the student's specifications are correct, will the egg break if the student drops the egg from shoulder height (5 ft) off a building 80 ft high? What is the maximum height the egg can be dropped from? (Note: The acceleration due to gravity is 32 ft/s^2.)

ANS:
The velocity can be calculated by using the height from which the egg is dropped. The height is 80 ft plus the height of the student, or 85 ft.

$$v = \sqrt{(v_0)^2 + 2ad}$$
$$= \sqrt{0 + 2(32)(85)}$$
$$= \sqrt{5440}$$
$$v = 73.76$$

Since the speed is less than 80 ft/s, the egg will not crack.
The maximum height is limited by a velocity of 80 ft/s, so

$$v = \sqrt{(v_0)^2 + 2ad}$$
$$80 = \sqrt{0 + 2(32)d}$$
$$80 = \sqrt{64d}$$
$$6400 = 64d$$
$$100 = d$$

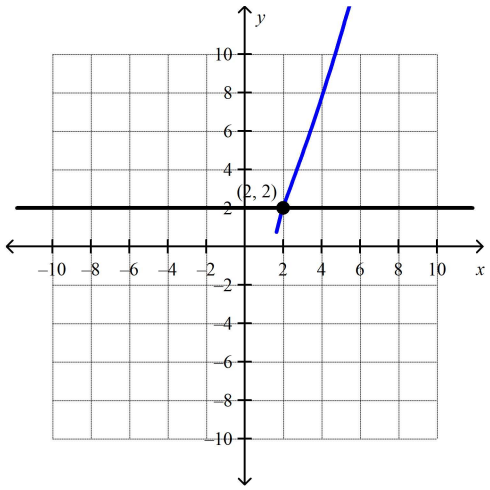
The maximum height is 100 ft.

DIF: Average OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

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12. Solve the equation $\sqrt{x^3 - 4} = 2$ graphically.

ANS:



DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: graphical solution

13. Solve the equation $\sqrt{x^3 - 7} = 1$ algebraically.

ANS:

$$\sqrt{x^3 - 7} = 1$$
$$x^3 - 7 = 1$$
$$x^3 = 8$$
$$x = 2$$

DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

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PROBLEM

1. The speed, s, in metres per second, of sound in dry air is can be described by the function $s = 331.3\sqrt{1 + \frac{T}{273.15}}$, where T is temperature, in degrees Celsius.

a) Determine the domain and range of the function.
b) Determine the speed of sound, to the nearest tenth of a metre per second, for each of the following temperatures.
i) –8°C
ii) 12°C
iii) 32°C
c) Graph the function and determine the x-intercept using the graph.
d) What is the meaning of the x-intercept in this context?

ANS:
a) Domain: $\{T \mid T \geq -273.15, T \in R\}$
Range: $\{s \mid s \geq 0, s \in R\}$

b) i) $s = 331.3\sqrt{1 + \frac{T}{273.15}}$

$= 331.3\sqrt{1 + \frac{-8}{273.15}}$

$s = 326.4$

ii) $s = 331.3\sqrt{1 + \frac{T}{273.15}}$

$= 331.3\sqrt{1 + \frac{12}{273.15}}$

$s = 338.5$

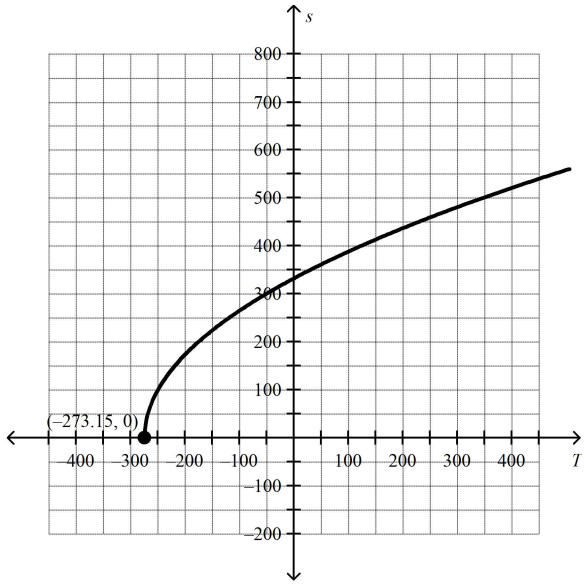
iii) $s = 331.3\sqrt{1 + \frac{T}{273.15}}$

$= 331.3\sqrt{1 + \frac{32}{273.15}}$

$s = 350.2$

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c)



The *x*-intercept is –273.15.

d) The *x*-intercept is the temperature when the speed of sound is 0.

DIF: Average OBJ: Section 2.1 | Section 2.3 NAT: RF13
TOP: Radical Functions and Transformations | Solving Radical Equations Graphically
KEY: x-intercepts | domain | range

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2. The manufacturer of a new Global Positioning Satellite (GPS) system wants to predict the consumer interest in its new device. The company uses the function $I(w) = -3\sqrt{-w-1} + 15$ to model the number, I, in thousands, of pre-orders for the GPS as a function of the number, w, of weeks before the GPS release date.

a) What are the domain and range and what do they mean in this situation?

b) Identify the transformations represented by the function compared to $y = \sqrt{w}$.

c) Graph the function and explain what the shape of the graph indicates about the situation.

d) Determine the number of pre-orders the manufacturer can expect to have 10 weeks before the release date.

ANS:

a) Since $I \geq 0$, solve $0 = -3\sqrt{-w-1} + 15$ to find the lower boundary of the domain.

$$0 = -3\sqrt{-w-1} + 15$$

$$3\sqrt{-w-1} = 15$$

$$\sqrt{-w-1} = 5$$

$$-w-1 = 25$$

$$-w = 26$$

$$w = -26$$

The upper boundary of the domain is the value of x at the vertex, (–1, 15). So the domain is $\{w | -26 \leq w \leq -1, w \in R\}$ and the range is $\{I | 0 \leq I \leq 15, I \in R\}$.

The domain is negative, indicating the weeks remaining before the launch of the GPS. The maximum number of GPS units expected to be pre-ordered is 15 thousand.

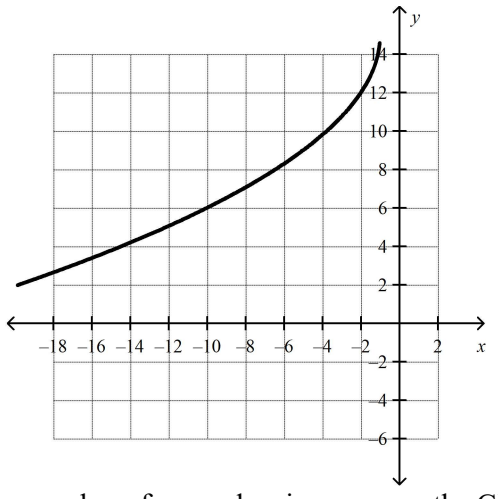
b) $a = -3$: reflection in the x-axis with a vertical stretch by a factor of 3

$b = -1$: reflection in the y-axis

$h = -1$: translation of 1 unit left

$k = 15$: vertical translation up 15 units

c)



The number of pre-orders increases as the GPS release date gets closer. Since w is negative, it represents the number of weeks remaining before release. The function's maximum of 15 thousand is the maximum number of pre-orders

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d) $I(w) = -3\sqrt{-w-1} + 15$

$I(10) = -3\sqrt{-(-10)-1} + 15$

$= -3\sqrt{10-1} + 15$

$= -3\sqrt{9} + 15$

$= -3(3) + 15$

$I(10) = 6$

Six thousand pre-orders can be expected 10 weeks before the release date.

DIF: Difficult OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: translations | graph | domain | range

3. Consider the function $f(x) = 2(x-1)^2 - 3$.

a) Determine the square root of the function.

b) Graph the functions on the same set of axes.

c) State the domains and ranges of the graphs.

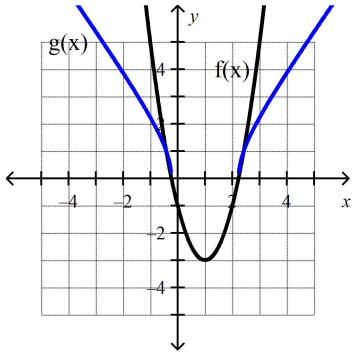
d) Describe the relationship between the domain and range of $f(x)$ and the domain and range of $g(x)$.

ANS:

a) $g(x) = \sqrt{f(x)}$

$g(x) = \sqrt{2(x-1)^2 - 3}$

b)



c) The domain of $f(x)$ is $\{x|x \in R\}$ and the domain of $g(x)$ is $\left\{x|x \le \left(1-\sqrt{\frac{3}{2}}\right), x \ge \left(1+\sqrt{\frac{3}{2}}\right), x \in R\right\}$. The range of $f(x)$ is $\{y|y \ge -3, y \in R\}$ and the range of $g(x)$ is $\{y|y \ge 0, y \in \mathbb{R}\}$.

d) The domain of $g(x)$ consists of all values of the domain of $f(x)$ where $f(x) \ge 0$. The range of $g(x)$ consists of the square root of all positive values of the range of $f(x)$.

DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically
KEY: graph | square root of a function | domain | range

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4. For $f(x) = 5\sqrt{-x}$ and $g(x) = -2\sqrt{6(x+2)} - 3$, do the following.

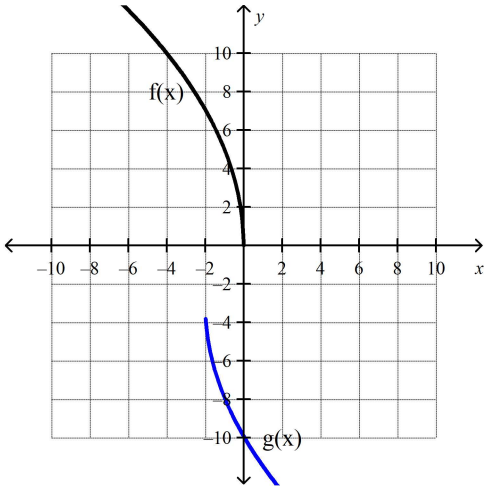
a) Graph $f(x)$ and $g(x)$ on the same set of axes.

b) Determine the domain and range of each function.

c) Explain which transformations would need to be applied to the graph of $f(x)$ to obtain the graph of $g(x)$.

ANS:

a)



b) $f(x)$: Domain: $\{x | x \le 0, x \in R\}$; Range: $\{y | y \ge 0, y \in R\}$

$g(x)$: Domain: $\{x | x \ge -2, x \in R\}$; Range: $\{y | y \le -3, y \in R\}$

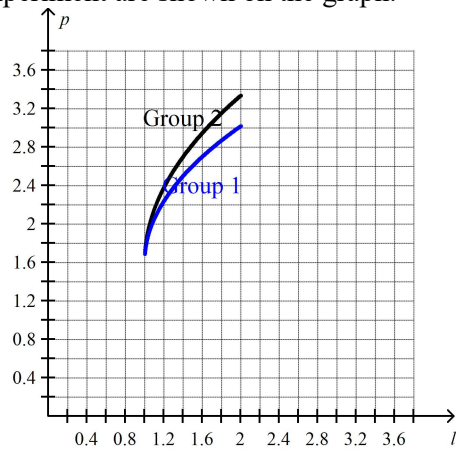
c) The vertical stretch changes from 5 to 2, which means a vertical compression by a factor of $\frac{2}{5}$.

There is a horizontal compression by a factor of $\frac{1}{6}$. The graph is reflected in both the x-axis and the y-axis. The graph is translated 2 units left and 3 units down.

DIF: Difficult OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations
KEY: graph | transformations | domain | range

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5. Two groups of students are conducting a lab to determine the relationship between the period, p, in seconds, of a pendulum and the length, l, in metres, of the string. The curves of best fit from the experiment are shown on the graph.



a) When asked the type of function that could be used to model their findings, both groups argue that a radical function can be used. Do you agree with each group?

b) How do these graphs differ from the graph of $f(x) = \sqrt{x}$?

c) Write a function to approximate the graph for each group.

d) What may have caused the differences in the data between the two groups? Justify your answer in terms of transformations.

ANS:

a) In both cases, the functions are in the shape of a transformed radical function.

b) In both cases, the function is stretched horizontally. The graph created by group 1 has a larger horizontal stretch than the graph created by group 2. Vertical and horizontal translations are applied to both graphs.

c) Answers may vary. Sample answer:

Group 1: $f(x) = \sqrt{2(x-1)} + 1.6$

Group 2: $g(x) = \sqrt{3(x-1)} + 1.6$

d) Answers may vary. Sample answer:

Group 1 may have pushed the pendulum or Group 2 may have started the timer early. Students should note that the only difference between the two groups can be accounted for by a horizontal stretch in the function.

DIF: Difficult OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: graph | transformations

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6. The kinetic energy (energy of motion), E, in joules, of an object is given by the equation $E = \frac{1}{2}mv^2$, where m represents the mass of the object, in kilograms, and v represents its speed, in metres per second.

a) Determine the general equation for the velocity of a mass as a function of its kinetic energy.

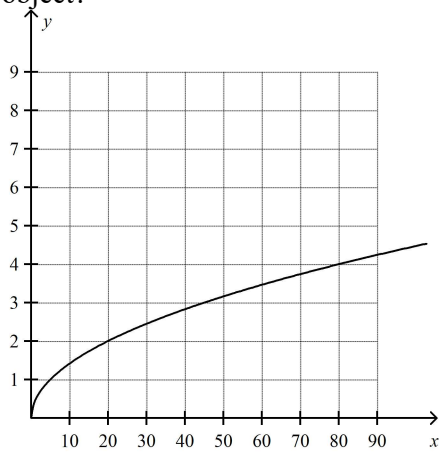
b) Find the speed of an object of mass 12 kg moving with a kinetic energy of

i) 200 J

ii) 420 J

c) Graph the function if the mass is 12 kg.

d) John conducts an experiment and graphs the data, resulting in the graph below. What is the mass of the object?



ANS:

a) $E = \frac{1}{2}mv^2$

$2E = mv^2$

$\frac{2E}{m} = v^2$

$v = \sqrt{\frac{2E}{m}}$

b) i) Substitute $m = 12$ and $E = 200$ in the equation.

$v = \sqrt{\frac{2E}{m}}$

$= \sqrt{\frac{2(200)}{12}}$

$v \approx 5.8$

The speed of the object is 5.8 m/s.

ii) Substitute $m = 12$ and $E = 420$ in the equation.

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$$v = \sqrt{\frac{2E}{m}}$$

$$= \sqrt{\frac{2(420)}{12}}$$

$v \approx 8.4$

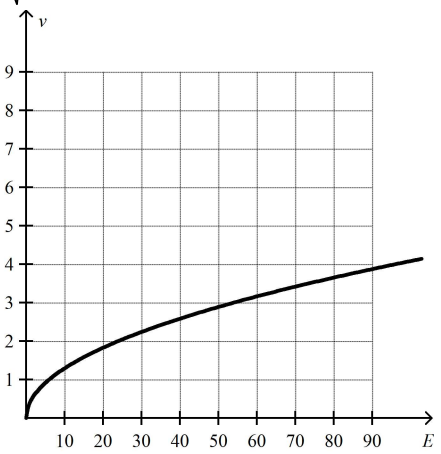
The speed of the object is 8.4 m/s.

c) When $m = 12$,

$$v = \sqrt{\frac{2E}{m}}$$

$$= \sqrt{\frac{2E}{12}}$$

$$v = \sqrt{\frac{E}{6}}$$



d) The point (20, 2) is on the graph. Substitute this into the equation $v = \sqrt{\frac{2E}{m}}$ and solve for m.

$$v = \sqrt{\frac{2E}{m}}$$

$$2 = \sqrt{\frac{2(20)}{m}}$$

$$4 = \frac{40}{m}$$

$m = 10$

The mass of the object is 10 kg.

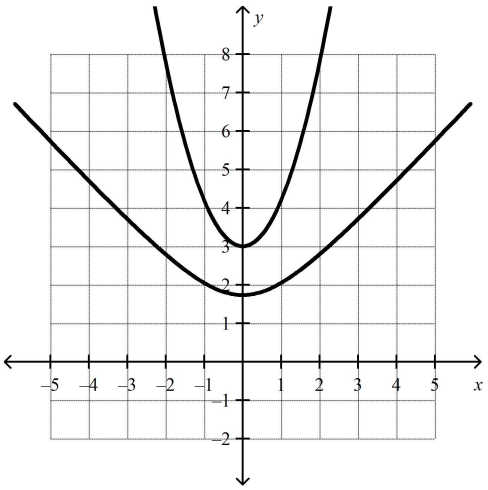
DIF: Average OBJ: Section 2.1 NAT: RF13
TOP: Radical Functions and Transformations KEY: graph | horizontal stretch

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7. Graph the functions below, and then identify and compare their domains and ranges.

$y_1 = 3 + 1.2x^2$ and $y_2 = \sqrt{3 + 1.2x^2}$

ANS:



$y_1 = 3 + 1.2x^2$ is a quadratic. The range of a positive quadratic function is all values of y so that y is greater than the minimum. In this case, the minimum is $y = 3$, so the range is $y \geq 3$. The domain is defined for all values of x, so the domain is all real numbers.

$y_2 = \sqrt{3 + 1.2x^2}$ is a radical function and is defined for all values of y greater than the minimum value at $x = 0$. Therefore, $y \geq \sqrt{3}$. The domain is defined for all values of x. Therefore, the domain is all real numbers.

DIF: Average OBJ: Section 2.2 NAT: RF13 TOP: Square Root of a Function
KEY: graph | domain | range | square root of a function

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8. Sun-Eui, a graduate student, is using the function $N(t) = 2.1\sqrt{t} + 0.6$ to model the number of flowers a new hybrid bee can pollinate. For the function, N represents the number, in thousands, of flowers pollinated and t represents the time, in days.

a) If there are 10 000 flowers, how long will it take to pollinate all of the flowers?

b) Sun-Eui believes that the second generation of bees can reduce the time to pollinate the flowers by half by changing the function to $N(t) = 1.05\sqrt{t} + 0.3$. Is she correct? Justify your reasoning.

c) Should Sun-Eui continue with the project?

ANS:

a)
$$N(t) = 2.1\sqrt{t} + 0.6$$
$$10 = 2.1\sqrt{t} + 0.6$$
$$10 - 0.6 = 2.1\sqrt{t}$$
$$\frac{9.4}{2.1} = \sqrt{t}$$
$$20.0 = t$$

It will take 20 days to pollinate the flowers.

b) Substitute $N = 10$ into the new function:
$$N(t) = 1.05\sqrt{t} + 0.3$$
$$10 = 1.05\sqrt{t} + 0.3$$
$$10 - 0.3 = 1.05\sqrt{t}$$
$$\frac{9.7}{1.05} = \sqrt{t}$$
$$85 = t$$

Sun-Eui is not correct since the new function will result in a time of 85 days, which means that it will take more than four times as long to pollinate the 10 000 flowers.

c) Sun-Eui should only continue with the project once she understands the math better so she can construct a function that will give half the pollination time.

DIF: Difficult OBJ: Section 2.3 NAT: RF13
TOP: Solving Radical Equations Graphically KEY: algebraic solution

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